

Pre- and postnatal development

PRENATAL STAGES

The absolute size of an embryo or fetus does not afford a reliable indication of either its chronological age or the stage of structural organization, even though graphs based on large numbers of observations have been constructed to provide averages. All such data suffer from the difficulty of timing the moment of conception in humans. It has long been customary to compute the age, whether in a normal birth or an abortion, from the first day of the last menstrual period of the mother but, as ovulation usually occurs near the fourteenth day of a menstrual cycle, this 'menstrual age' is an overestimate of about 2 weeks. Where a single coitus can be held to be responsible for conception, a 'coital age' can be established, and the 'fertilization age' cannot be much less than this because of the limited viability of both gametes. It is usually held that the difference may be several days, which is a highly significant interval in the earlier stages of embryonic development. Even if the time of ovulation and coitus were known in instances of spontaneous abortion, not only would some uncertainty still persist with regard to the time of fertilization, but also there would remain an indefinable period between the cessation of development and the actual recovery of the conceptus.

To overcome these difficulties, early embryos have been graded or classified into developmental stages or 'horizons', on the basis of both internal and external features. The study of the Carnegie collection of embryos by Streeter (1942, 1945, 1948), and the continuation of this work by O'Rahilly and Müller (1987), provided, and continues to provide, a sound foundation for embryonic study and a means of comparing stages of human development with those of the animals routinely used for experimental study, namely: the chick, mouse and rat. Recent use of ultrasound for the examination of human embryos and fetuses *in utero* has confirmed much of the staging data.

The development of a human from fertilization to birth is divided into two periods: embryonic and fetal. The embryonic period has been defined by Streeter as 8 weeks post fertilization, or 56 days. This timescale is divided into 23 Carnegie stages, a term introduced by O'Rahilly and Müller (1987) to replace developmental 'horizons'. The designation of stage is based on external and internal morphological criteria and not on length or age.

Embryonic staging

Embryonic stages 1–10 are shown in detail in Figure 8.1. Estimations of embryonic length may be 1–5 mm less than equivalent *in vivo* estimates, reflecting the shrinkage caused by the fixation procedures that are inevitably used in embryological studies. O'Rahilly and Müller (2000, 2010) have revised some of the ages that were previously assigned to early embryonic stages, pointing out that interembryonic variation may be greater than had been thought and that, consequently, some ages may have been underestimated. They note that, as a guide, the age of an embryo can reasonably be estimated from the embryonic length within the range 3–30 mm, by adding 29 to the length. Correlating the age of any stage of development to an approximate day may be unreliable, and a generalization using the number of weeks of development might now be more appropriate.

The stages of development encompass all aspects of internal and external morphogenetic change that occur within the embryo within the duration of the stage. They are used to convey a snapshot of the status of the development of all body systems within a particular time-frame. Figure 14.1 shows the external appearance of embryos from stage 6 to stage 23, with details of their size and age in days. The correlation of external appearance of the embryo with internal development is shown in Figure 14.2.

Obvious external features provide some guidance to the changes occurring within embryos during successive stages. Somite number is

related to early embryonic stages, and once the number of somites is too great to count with accuracy, the degree of development of the pharyngeal arches is often used. External staging becomes more obvious when the limb buds appear. The upper limb bud is clearly visible at stage 13 and, by stage 16, the acquisition of a distal paddle on the upper limb bud is characteristic. At stage 18, the lower limb bud now has a distal paddle, whereas the upper limb bud has digit rays that are beginning to separate. By stage 23, the embryo has a head that is almost erect and rounded, and eyelids are beginning to form. The limbs look far more in proportion and fingers and toes are separate. At this stage, the external genitalia are well developed, although they may not be sufficiently developed for the accurate determination of the sex.

Historically, the onset of bone marrow formation in the humerus was used by Streeter to indicate the end of the embryonic and the beginning of the fetal period of prenatal life. The fetal period occupies the remainder of intrauterine life; growth is accentuated, although differentiative processes continue up to and beyond birth. Overall, the fetus increases in length from 30 mm to 500 mm, and increases in weight from 2–3 g to more than 3000 g.

Fetal staging

Currently, there is no satisfactory system of morphological staging of the fetal period of development, and the terminology used to describe this time period reflects this confusion. The terms 'gestation', 'gestational age' and 'gestational weeks' are considered ambiguous by O'Rahilly and Müller (2000), who recommend that they should be avoided. However, they are widely used colloquially within obstetric practice. Staging of fetal development and growth is based on an estimate of the duration of a pregnancy. Whereas development of a human from fertilization to full term averages 266 days, or 9.5 lunar months (28 day units), the start of pregnancy is traditionally determined clinically by counting days from the last menstrual period; estimated in this manner, pregnancy averages 280 days, or 10 lunar months (40 weeks). Figure 14.3 shows the embryonic timescale used in all descriptions of embryonic development and the obstetric timescale used to gauge the stage of pregnancy.

The predicted date of full term and delivery is revised after routine ultrasound examination of the fetus. Early ultrasound estimation of gestation increases the rate of reported preterm delivery (delivery at <37 weeks) compared with estimation based on the date of the last menstrual period (Yang et al 2002), possibly because delayed ovulation is more frequent than early ovulation; the predicted age of a fetus estimated from the date of the last menstrual period may differ by more than 2 weeks from estimates of postfertilization days.

Obstetric staging

In obstetric practice, the duration of the period of gestation is regarded as 9 calendar months, which is approximately 270 days. The period of pregnancy is divided into thirds, termed trimesters. The first and second trimesters each cover a period of 12 weeks, and the third trimester covers the period from 24 weeks to delivery. Although the expected date of delivery is computed at 40 weeks of pregnancy, the term of the pregnancy, i.e. its completion resulting in delivery, is considered normal between 37 and 42 weeks. Neonates delivered before 37 weeks are called preterm (or premature); those delivered after 42 weeks are post-term. The period from the end of week 24 and up to 7 days after birth is termed the perinatal period. Fetuses that are delivered and die before 24 weeks are considered to be miscarriages of pregnancy, although technological advances in neonatal care can now assist the delivery and support of infants younger than 24 weeks. Infants born after 24 weeks of pregnancy who subsequently die are classed as stillborn and contribute to the statistics of perinatal mortality. Studies that discuss fetal

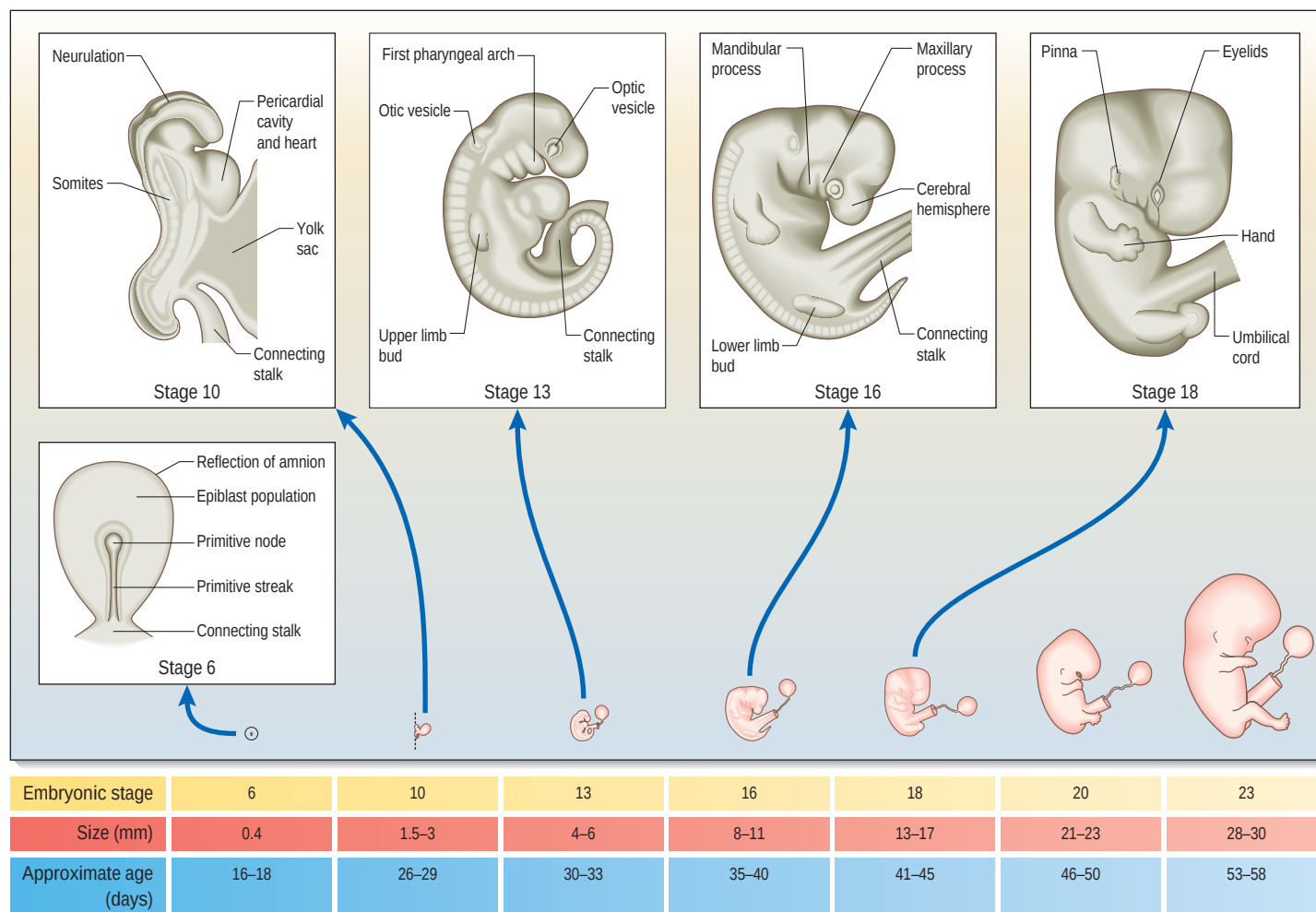


Fig. 14.1 The external appearance and size of embryos between stages 6 and 23. Early in development, external features are used to describe the stage, e.g. somites, pharyngeal arches or limb buds. (Adapted with permission from Rodeck CH, Whittle MJ 1999 *Fetal Medicine*. London: Churchill Livingstone.)

development and the gestational age of neonates, particularly those born before 40 weeks' gestation, use the obstetric estimated stages and age (menstrual age), unless they specifically correct for this. If a fetal ageing system is used, it is important to remember that the age of the fetus may be 2 weeks more than a comparable fetus that has been aged from postovulatory days.

Ultrasound staging

The difficulty of correlating the appearance of a chorionic sac, embryo or fetus on an ultrasound scan with age during the first trimester is related to the specificity of reporting the age. An age reported as within a week of pregnancy, e.g. week 12, will cover a period of 6 days (12 weeks 0 days up to 12 weeks 6 days). Therefore, it is recommended that sonographic estimation of age should be given as menstrual weeks and days (i.e. 12 weeks indicates 12 weeks 0 days) (Galan et al 2008).

First trimester scan An early ultrasound scan will detect implantation and viability of the embryo once a heart beat is detected, confirm multiple pregnancy and estimate the date of delivery. Cardiac activity can be identified by the sixth menstrual week (Galan et al 2008). Crown-rump length measurement is the most accurate predictor of menstrual age of the embryo during the first trimester (Galan et al 2008). Nuchal translucency is measured between 10 and 14 weeks to diagnose trisomy 21. The development of three-dimensional ultrasound scanning has enabled early detection of many anomalies previously diagnosed in the second trimester, including anencephaly, hydrocephalus and encephalocele (Pretorius et al 2014). Normograms of fetal spine growth have been constructed for a Taiwan population, which show mean spine length increasing linearly between 11 and 14 weeks (Cheng et al 2010). The study also noted a tendency for spinal extension as early as 11 weeks.

Second trimester scan Routine scanning at 18–20 weeks (obstetric staging) is used to confirm the delivery date, and assess not only the position of the placenta but also the presence of fetal anomalies that would require special intervention following delivery, such as cardiac

defects, lung immaturity, renal agenesis, polyhydramnios and defects of the anterior abdominal wall (gastroschisis and exomphalos). Estimations are made of the following: biparietal skull diameter, taken through a plane of section that traverses the third ventricle and thalami; head circumference through a plane that traverses the third ventricle and thalami plus the cavum septi pellucidi anteriorly and the tentorial hiatus posteriorly; abdominal circumference through a plane where the transverse diameter of the liver is greatest, the appearance of the lower ribs is symmetrical and the junction of the left and right portal veins is identified; and femoral length, which measures the ossified portions of the diaphysis and metaphysis (Fig. 14.4).

Between 15 and 28 weeks, the biparietal diameter is the most accurate index of fetal menstrual age and the expected date of delivery. Other measurements that may also be taken include transverse cerebellar diameter and foot length. The amount and type of fetal movement, breathing movements and visceral functions, exemplified by bladder emptying, peristaltic action and colonic echogenicity, are noted. Three- and four-dimensional ultrasound scans are now routinely constructed (Video 14.1).

Gender can be identified later in development but this information is not always routinely passed on to parents.

Constructions of ultrasound biometry charts for fetal aging now take into account the ethnic population under consideration; it is recommended that locally developed charts specific to the population should be used. The use of these charts means that factors that may influence fetal biometry, including maternal age and nutritional status, maternal height, weight, parity and smoking habits, are noted, facilitating accurate prediction of small-for-date and growth-retarded fetuses. The World Health Organization (WHO) has published a protocol for the collection of data for a multicentre study of fetal growth (Merialdi et al 2014). INTERGROWTH-21st, an international, multicentre project, also aims to standardize protocols for collecting anthropometric measurements and constructing charts describing optimal fetal and preterm postnatal growth (Sarris et al 2013, Uauy et al 2013).

An outcome of routine ultrasound examination of embryos and fetuses for anomalies may be a change to the perinatal management,

Embryonic stages	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Weeks post ovulation	1	2	3	4	5	6	7	8	9	10	11	12						
External appearance			Head and tail folding Pharyngeal arches			Upper lip Palate		Digits on hand External ear	Eyelids fuse									
Nervous			Neurulation First neural crest cells		Otic vesicle Optic cup	Anterior lobe pituitary		Posterior lobe pituitary Membranous labyrinth										
Respiratory			Trachea Lung buds			Primary bronchi		Further division of bronchi										
Gastrointestinal			Fore-, mid-, hindgut Urorectal septum	Thyroid Liver		Pharyngeal pouches dorsal and ventral Pancreas Rotation of stomach		Midgut loop rotating									Midgut loop returns to abdomen	
Urinary			Mesonephros Mesonephric duct Ureteric bud			Metanephric nephrons Major calyces		Minor calyces	Kidneys ascend									
Reproductive			Germ cells in allantois wall Indifferent gonad			Müllerian ducts Testis differentiating External genitalia indifferent			Uterus and uterine tubes Vagina								Testis at inguinal canal Prostate External genitalia differentiating	
Cardiovascular			Primitive vascular system Heart tube	Septum primum Heart beats		Septation of ventricles Spleen		Septum secundum										
Musculoskeletal			Somite period 20 days 30 days Forelimb bud			Forelimb digit rays Hindlimb bud		Cartilaginous part of skull		Membranous part of skull								

Fig. 14.2 A timetable of development of the body systems. The development of individual systems can be seen progressing from left to right. Embryonic stages and weeks of development are shown. Embryonic stages are associated with external and internal morphological features rather than embryonic length. To identify the systems and organs at risk at any time of development, follow a vertical progression from top to bottom.

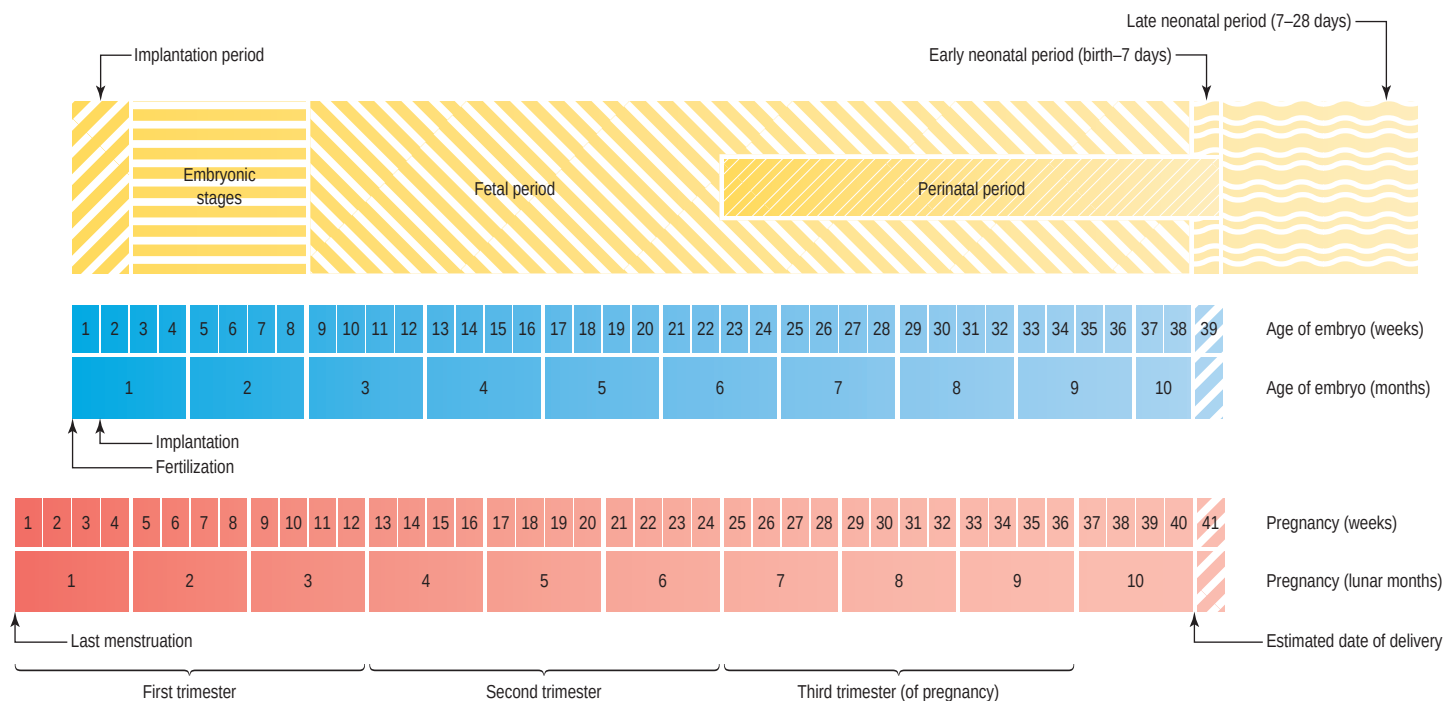
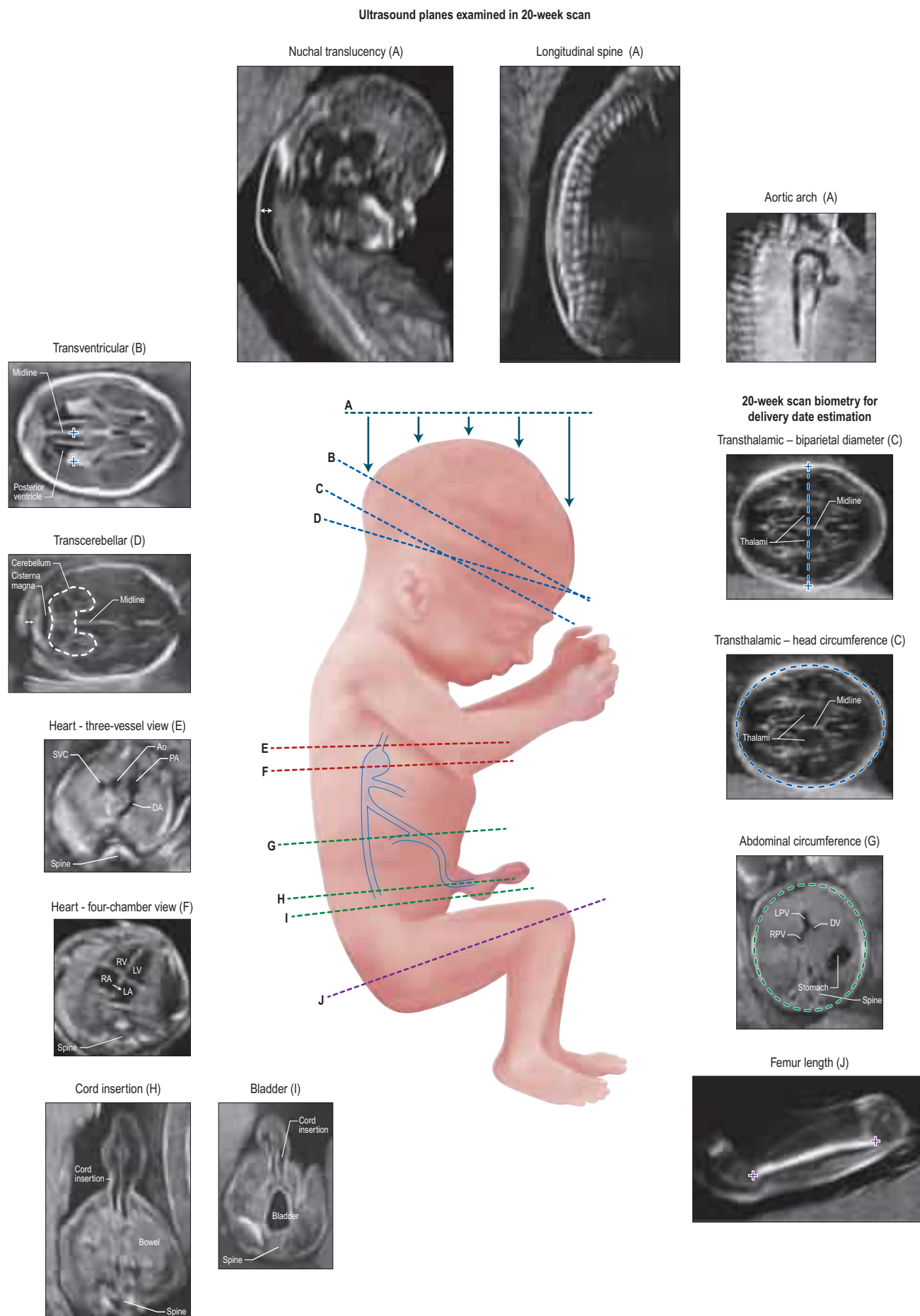


Fig. 14.3 The two timescales used to depict human development. Embryonic development, in the upper scale, is counted from fertilization (or from ovulation, i.e. in postovulatory days; see O'Rahilly and Müller (1987)). Throughout this book, times given for development are based on this scale. The clinical estimation of pregnancy is counted from the last menstrual period and is shown on the lower scale; throughout this book, fetal ages relating to neonatal anatomy and growth will have been derived from the lower scale. Note that there is a 2-week discrepancy between these scales. The perinatal period is very long because it includes all preterm deliveries.



i.e. to the time, method and place of delivery of the fetus, or the parents may choose to terminate the pregnancy to minimize concerns about fetal and neonatal suffering and long-term disability. Termination may be chosen for severe, untreatable inherited metabolic disorders (e.g. Tay-Sachs disease), severe chromosomal anomalies (e.g. trisomy 13), lethal bone dysplasias, lethal anomalies such as anencephaly and other extreme neurological defects, and bilateral renal agenesis (Cass 2011). Improvements in prenatal screening and diagnosis have led to an overall increase in the prevalence of reported birth defects and overall lower perinatal mortality rates, reflecting increased early terminations of pregnancy (Cass 2011).

Magnetic resonance imaging

Advances in prenatal ultrafast magnetic resonance imaging (MRI) now provide further ways to detect fetal anomalies, capturing particularly clear images (Sepulveda et al 2012) (Fig. 14.5). MRI is considered a useful adjunct to ultrasound imaging at 20–22 weeks because it enables better management planning for known or suspected anomalies (Reddy et al 2014).

Fetal development

Although accurate morphological stages are not available for the fetal period, the developmental progression is broadly clear. During the fourth and fifth months, the fetus has a head and upper limbs that are

still disproportionately large. (Although the rates of growth of the trunk and lower limbs increase during the remainder of intrauterine life, the disproportion is present after birth and, to a diminishing degree, is retained throughout childhood and on into the years of puberty.) A covering of primary hair, lanugo, appears. Towards the end of this period, sebaceous glands become active; the sebum that is secreted blends with desquamated epidermal cells to form a cheesy covering over the skin, the vernix caseosa, which is usually considered to protect the skin from maceration by the amniotic fluid. At about this time, the mother becomes conscious of fetal movement, formerly termed 'quickening'.

In the sixth month, the lanugo darkens, the vernix caseosa is more abundant and the skin becomes markedly wrinkled. The eyelids and eyebrows are now well developed. During the seventh month, the hair of the scalp is lengthening and the eyebrow hairs and the eyelashes are well developed. The eyelids themselves separate and the pupillary membrane disappears. The body becomes more plump and rounded in contour and the skin loses its wrinkled appearance as a result of the increased deposition of subcutaneous fat. Fetal length has increased to approximately 350 mm and weight to about 1.5 kg. Towards the end of this month, the fetus is viable; if born prematurely, it may be able to survive without the technological assistance found in neonatal intensive care units and its postnatal development can proceed normally.

Throughout the remaining lunar months of normal gestation, the covering of vernix caseosa is prominent. There is a progressive loss of lanugo, except for the hairs on the eyelids, eyebrows and scalp. The bodily shape becomes more infantile but, despite some acceleration in their growth, the legs have not quite equalled the arms in length proportionately, even at the time of birth. The thorax broadens relative to the head, and the infra-umbilical abdominal wall shows a relative increase in area, so that the umbilicus gradually becomes more centrally situated. Average lengths and weights for the eighth, ninth and tenth months are 40, 45 and 50 cm and 2, 2.5 and 3–3.5 kg, respectively. The rate of fetal growth slows from 36 to 40 weeks in response to the physical limitation imposed by the maternal uterus. Birth weight thus reflects the maternal environment more than the genotype of the child. This slowing of the growth rate enables a genetically larger child developing within a small mother to be delivered successfully. After birth, the growth rate of the neonate increases; the rate of weight gain reaches a peak some 2 months postnatally.

Just before birth, the lanugo almost disappears and the umbilicus is central. The testes, which begin to descend with the processus vaginalis of peritoneum during the seventh month and are approaching the scrotum in the ninth month, are usually scrotal in position. The ovaries are not yet in their final position at birth; although they have attained their final relationship to the uterine folds, they are still above the level of the pelvic brim.

Fetal surgery

The use of routine ultrasound examination and MRI of fetuses has led to the development of a number of prenatal surgical interventions: e.g. to correct placental or membrane anomalies resulting in twin-to-twin transfusion and the production of amniotic bands (Deprest et al 2014); or to offer improved outcomes for meningomyelocele on the basis that the neural tissue may become secondarily damaged due to exposure to amniotic fluid and mechanical traumatic injury during gestation (Adzick 2013, Danzer and Johnson 2014, Cohen et al 2014).

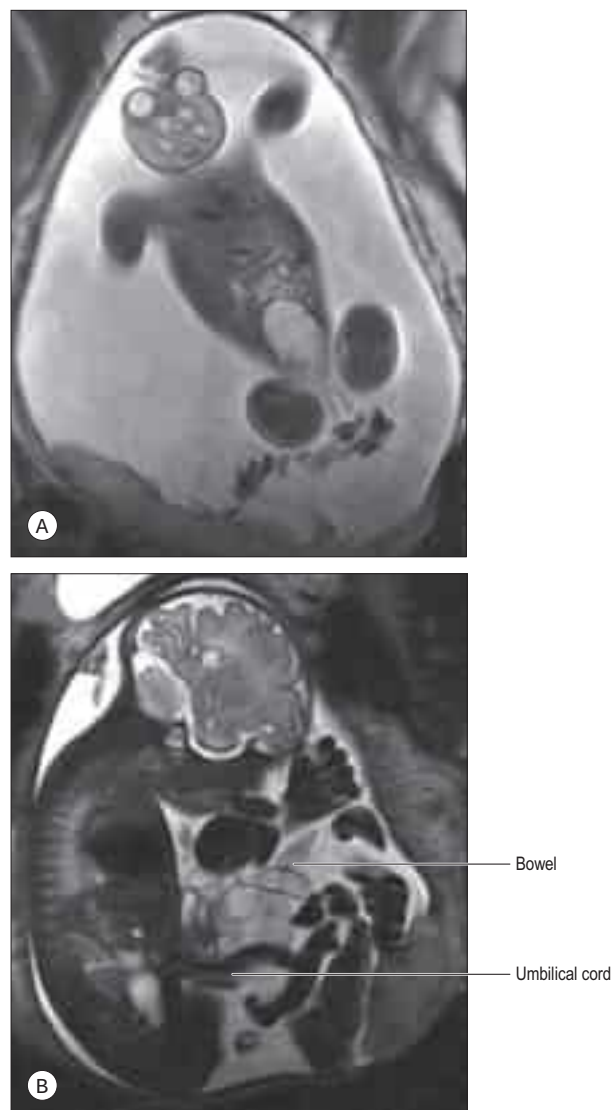


Fig. 14.5 MRI scans of fetal anomalies. **A**, Polyhydramnios may arise as a result of oesophageal compression, impaired swallowing and lack of absorption of amniotic fluid from the gut. **B**, Gastroschisis with loops of bowel in the extracellular coelom (chorionic cavity) and normal insertion of the umbilical cord. (With permission from Sepulveda W, Ximenes R, Wong AE, et al 2012 Fetal magnetic resonance imaging and three-dimensional ultrasound in clinical practice: Applications in prenatal diagnosis. *Best Pract Res Clin Obstet Gynaecol* 26:593–624.)

GROWTH IN UTERO

Alterations in the availability of nutrients to the fetus at particular stages of pregnancy elicit adaptive responses by the fetus. These may ensure fetal coping but may also result in pathology in adult life; the nutritional status of pregnant females is therefore of fundamental importance for the health of the next generation. Under- or overnutrition during fetal life may lead to metabolic changes in childhood, adolescence and adulthood, and pass to successive generations. Poor nutrition at critical stages of fetal life permanently alters the normal developmental pattern of a range of organs and tissues, e.g. the endocrine pancreas, liver and blood vessels, resulting in their pathological responses to certain conditions in later adult life (Barker et al 1993, Catalano and Hauguel-De Mouzon 2011). Maternal hyperglycaemia leads to changes in fetal metabolism and pathology in the child and adult (Pedersen 1952).

An increase in placental size occurs in pregnancy as an adaptive response to both high altitude and mild undernutrition, particularly

Other conditions have also been treated prenatally with limited improvement in mortality. Thus, congenital diaphragmatic hernia may be treated by percutaneous fetoscopic endoluminal tracheal occlusion from 25 to 33 weeks' gestation to prevent loss of lung fluid and enhance lung growth ([Deprest et al 2004](#)). Fetal airway patency has also been preserved during birth through perinatal *ex utero* intrapartum (EXIT) treatment, where a portion of the fetus is delivered through a hysterectomy incision for surgery while the fetus remains attached to the utero-placental circulation. After the airway procedure is completed, the fetus is completely delivered ([Deprest et al 2014](#)).

PRE- AND POSTNATAL DEVELOPMENT

during mid-pregnancy. However, although a larger placenta may be better able to deliver the full nutritional requirements of the fetus, the perfusion of a larger placenta is not without problems. It may produce changes in fetal blood flow and placental enzymes, and in the normal structure of the fetal vessel wall or of its responses to circulating trophins, e.g. catecholamines or angiotensin II, which will continue into adult life. Undernutrition in later pregnancy does not produce the same sequelae and placental enlargement does not occur. However, fetal growth slows and fetal wasting may occur as oxygen, glucose and amino acids are redistributed to the placenta to maintain its function.

Transition to extrauterine life

The type of birth confers developmental outcomes on the neonate that affect its subsequent growth. Caesarean section for preterm delivery appears not to be associated with improved neonatal outcomes but is associated with increased risk of respiratory distress syndrome due to slower postnatal movement of fluid out of the lungs (Werner et al 2012, Bhatta and Keriakos 2011, Bellini et al 2006). The dramatic rise of this delivery method is not necessarily driven by improved neonatal outcomes because caesarean section has been shown to increase the odds of developing asthma and type 1 diabetes in later life by 20%. Meta-analyses indicate that the endocrine milieu during caesarean section may lead to epigenetic effects on hepatic and metabolic function and so affect immune function in adult life, leading, in some instances, to food allergy and obesity (Steer and Modi 2009, Hyde and Modi 2012, Song et al 2013).

Prior to birth, the gut is exposed to hundreds of proteins, metabolites and cytokines that are swallowed in amniotic fluid; there is some evidence for the presence of bacterial species in amniotic fluid prior to delivery. Establishment of the gut microbiota by exposure to maternal vaginal and colonic bacteria during vaginal delivery leads to normal maturation of the gut wall. Gut colonization in preterm infants is delayed (Cilieborg et al 2012). Infants born via caesarean section may have primary gut flora disturbance for up to 6 months after birth and associated delay in postnatal immune development (Neu and Rushing 2011, Neu and Mai 2012, Matamoros et al 2013).

The establishment of enteral feeding has profound effects on early postnatal maturation. Human breast milk contains many bioactive substances that enhance gut maturity. It improves gastric emptying, encourages the growth of the microbiotome and contains a range of proteins, including large amounts of secretory immunoglobulin A and cytokines (e.g. interleukin (IL)-10) (Jakaitis and Denning 2014). Feeding preterm infants human milk is associated with less feeding intolerance and is thought to provide protection against diabetes and obesity in later life (Valentine and Morrow 2012). The rate of growth of infants fed breast milk for the first 6 months of postnatal life is different to that of formula-fed infants. Delivery of low-weight preterm babies, followed by parenteral feeding with no enteral nutrition, disrupts normal gut maturation and may lead to necrotizing enterocolitis (Wynn and Neu 2012). However, a balance with the gut microbiome is necessary; although full-term neonates born in tropical countries have similar villous height to those born in temperate climates, villous length in the small intestine shortens within 2–12 months of birth as a consequence of tropical enteropathy (Ramakrishna et al 2006).

The time of transition to extrauterine life is permanently recorded in the primary teeth and can be reliably demonstrated in forensic dental identification. The circadian growth rhythm of dentine and enamel deposition in tooth germs is temporarily blocked by the metabolic stress of delivery; the resultant change in enamel prism deposition is seen as a neonatal line or ring that can be detected in all deciduous teeth and permanent first molar teeth of live births. The thickness of the line is related to birth difficulties, being thinner in caesarean section and thicker in vaginal delivery (Sabel et al 2008, Canturk et al 2014).

NEONATE

The neonatal period extends from birth to 28 days postnatally; it is divided into an early neonatal period from birth to 7 days, and a late neonatal period from 7 to 28 days. Technological advances have enabled successful management of preterm infants, many at ages that were considered non-viable a decade or two previously. Maturation processes involving local interactions and pattern formation still drive development at local and body-system levels in preterm infants. The sudden release of such fetuses into a gaseous environment of varying temperature, with full gravity and a range of microorganisms, promotes

the rapid maturation of some systems and the compensatory growth of others (in terms of responses to the effect of gravity or enteral feeding or exposure to microorganisms).

Details of the relative positions of the viscera and the skeleton in a full-term neonate are shown in Figures 14.6–14.8. The newborn infant is not a miniature adult, and extremely preterm infants are not the same as full-term infants. There are immense differences in the relations of some structures between the full-term neonate, child and adult, and there are also major differences between the 20-week-gestation fetus and the 40-week fetus, just before birth. The study of fetal anatomy at 20, 25, 30 and 35 weeks is vital for the investigative and life-saving procedures carried out on preterm infants today.

Neonatal measurements and period of time in utero

The tenth to ninetieth centile ranges for length of full-term neonates are 48–53 cm (Fig. 14.9A). Length of the newborn is measured from crown to heel. *In utero*, length has been estimated either from crown–rump length, i.e. the greatest distance between the vertex of the skull and the ischial tuberosities, with the fetus in the natural curved position, or from the greatest length exclusive of the lower limbs. Greatest length is independent of fixed points and thus much simpler to measure; it is generally taken to be the sitting height in postnatal life, and is the measurement recommended by O’Rahilly and Müller (2000) as the standard in ultrasound examination.

At birth, weight reflects the maternal environment, the number of conceptuses, the sex of the baby and the parity of the mother. Generally, full-term female babies are lighter than full-term males, twins are lighter than singletons, and later children tend to be heavier than the first-born. The tenth to ninetieth centile ranges for the weight of a full-term infant at parturition are 2700–3800 g (Fig. 14.9B), the average being 3400 g; 75–80% of this weight is body water and a further 15–28% is composed of adipose tissue.

Birth weight is noted against charts appropriate for ethnicity and categorized as low, normal and high. Low birth weight has been defined as less than 2500 g, very low birth weight as less than 1500 g, and extremely low birth weight as less than 1000 g. Infants may weigh less than 2500 g but not be premature by gestational age. Measurement of the range of weights that fetuses may attain before birth has led to the production of weight charts, which allow babies to be described according to how appropriate their birth weight is for their gestational age, e.g. small for gestational age, appropriate for gestational age or large for gestational age (Fig. 14.10). Small-for-gestational-age infants, also termed ‘small-for-dates’, are often the outcome of intrauterine growth retardation.

For both premature and growth-retarded infants, an assessment of gestational age, which correlates closely with the stage of maturity, is desirable. Gestational age at birth is predicted by its proximity to the estimated date of delivery and the results of ultrasonographic examinations during pregnancy. It is currently assessed in the neonate by evaluation of a number of external physical and neuromuscular signs. Scoring of these signs results in a cumulative score of maturity that is usually within ± 2 weeks of the true age of the infant. The scoring scheme has been devised and improved over many years.

Estimation of large-for-dates infants is based on assessment of fetal weight through ultrasound evaluation and some biometrical indices. Assessment of anterior abdominal wall width is thought to predict large-for-gestational-age babies (Walsh and McAuliffe 2012). Fetal macrosomia is defined as an absolute birth weight greater than 4000 g, 4500 g or 5000 g, or as a customized birth weight centile greater than the ninetieth, ninety-fifth or ninety-seventh centile for the infant’s gestation age (Walsh and McAuliffe 2012, Schwartz et al 2014). The precise definition may not necessarily be helpful, as some at-risk infants, not identified as large for dates from growth curve charts, might go unrecognized (Larma and Landon 2011). There is a correlation between macrosomia and short maternal stature; macrosomic fetuses are at risk of shoulder dystocia and brachial plexus injuries during vaginal delivery.

GROWTH IN INFANCY AND CHILDHOOD

After birth, there is a general decrease in the total body water but a relative increase in intracellular fluid. Normally, the newborn loses about 10% of its birth weight by 3–4 days postnatally because of loss of excess extracellular fluid. By 1 year, total body water makes up 60% of the body weight.

Animal studies have verified that maternal starvation during pregnancy decreases fetal intrauterine growth factor (IGF)-I concentrations and, together with a general hypoglycaemia, impairs the development of the fetal β cells of the pancreas. Moreover, fetal undernutrition induces insulin resistance in the tissues. The coexistence of insulin resistance and impaired β -cell development in the fetus appears to be important in the pathogenesis of type 2 diabetes. The risk of developing this type of diabetes is greatest in those individuals with low weight at birth and at 1 year, and who become obese as adults, thus challenging an already impaired glucose–insulin metabolism. Fetal IGF-I concentrations are also lower in infants who are short at birth as a result of a long period of maternal undernutrition; these individuals have an exaggerated response to growth hormone-releasing factor, which, together with low IGF-I concentrations, suggests a degree of growth hormone resistance.

It is now thought that the balance of hormonal environment in intrauterine and early postnatal life is necessary for future adult health. The presence of altered concentrations of hormones during critical periods of development may act as endogenous functional teratogens (Plagemann 2004).

Different birth phenotypes have been correlated with different pathological sequelae. Infants who are thin at birth, with a low ponderal index (weight/length³), tend to develop a combination of insulin resistance, hypertension, type 2 diabetes and lipid disorders, whereas those who are short in relation to head size tend to develop hypertension and high plasma fibrinogen concentrations. These associations have been reported in babies born small for dates, rather than in those born prematurely. Some babies of average weight also develop cardiovascular pathology; they are either thin at birth and small in relation to the size of their placenta, or of average weight but short in relation to head size and have below average weight gain during the first year (Barker et al 1993). For more recent views on this concept, the reader is directed to consult Godfrey and Barker (2000), Ross and Beal (2008), and Kelishadi and Poursafa (2014).

Obesity has become increasingly prevalent since the end of the twentieth century, leading to an increase in type 2 diabetes, hypertension, hyperlipidaemia, atherosclerosis and inflammation in later life. In pregnancy, an observed decrease in insulin sensitivity as pregnancy advances leads to increased nutrient transfer to the fetus. In response, the fetus increases insulin secretion, stimulates IGF-I production, and fetal growth and fat deposition are promoted (Vrachnis et al 2012). Obese females start their pregnancy with greater insulin resistance than females of normal weight, which means that their fetuses produce more IGF-I and continue to grow. The risk of fetal macrosomia is three times higher in females with poorly controlled gestational diabetes because their increased weight reflects fat mass rather than lean body mass (Catalano and Hauguel-De Mouzon 2011). Adiposity at birth is related to obesity and metabolic dysfunction in childhood, which may be perpetuated through adulthood in an ongoing cycle through the generations (Catalano and Hauguel-De Mouzon 2011, Poston 2012). A range of risk factors for pre-, peri- and postnatal issues in maternal obesity is given in McGuire et al (2010).

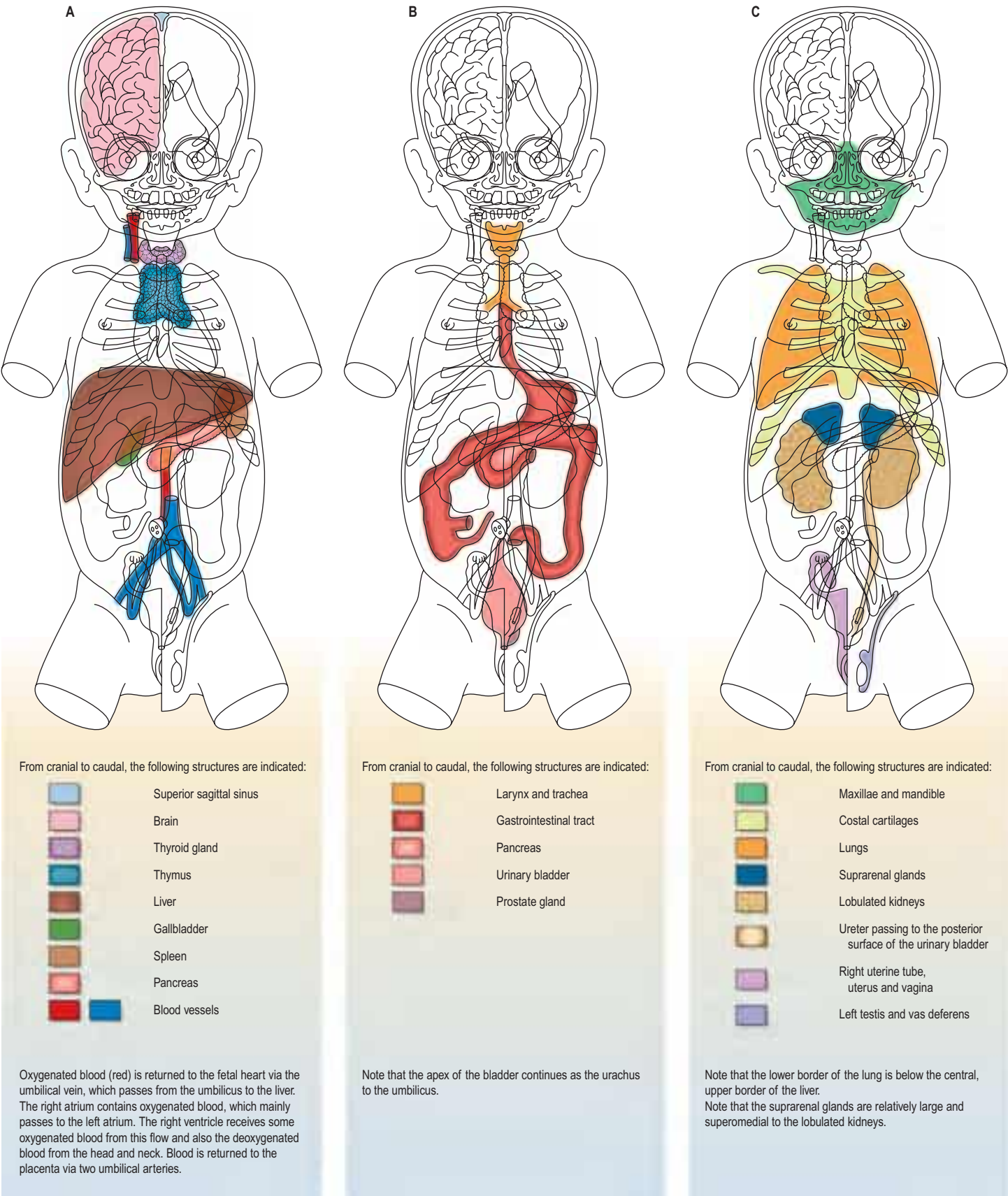


Fig. 14.6 A–C, Topographical representations of the anatomy of a full-term neonate. The surface markings of all organs are shown, with some coloured and others only in outline. The female genital tract is shown on the right of the body, and the male tract is shown on the left.

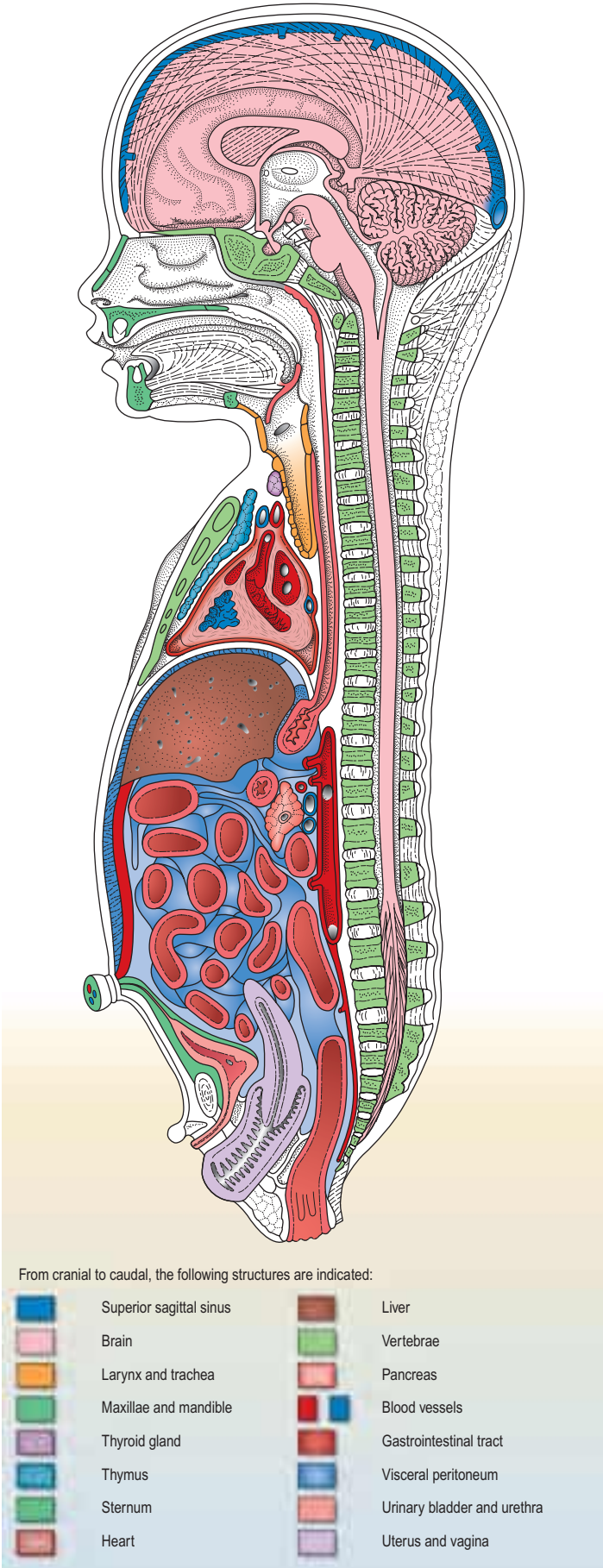


Fig. 14.7 A topographical representation of the anatomy of a full-term female neonate. The cut surfaces of the organs are the same colours as in Figure 14.6.



Fig. 14.8 The extent of the ossified skeleton in a full-term neonate. Note the derivation of the parts of the skeleton: the skull is derived from paraxial mesenchyme and neural crest mesenchyme; the axial skeleton, vertebrae and ribs are derived from paraxial mesenchyme; and the skeletal elements in the limbs are derived from somatopleuric mesenchyme, which forms the limb buds.

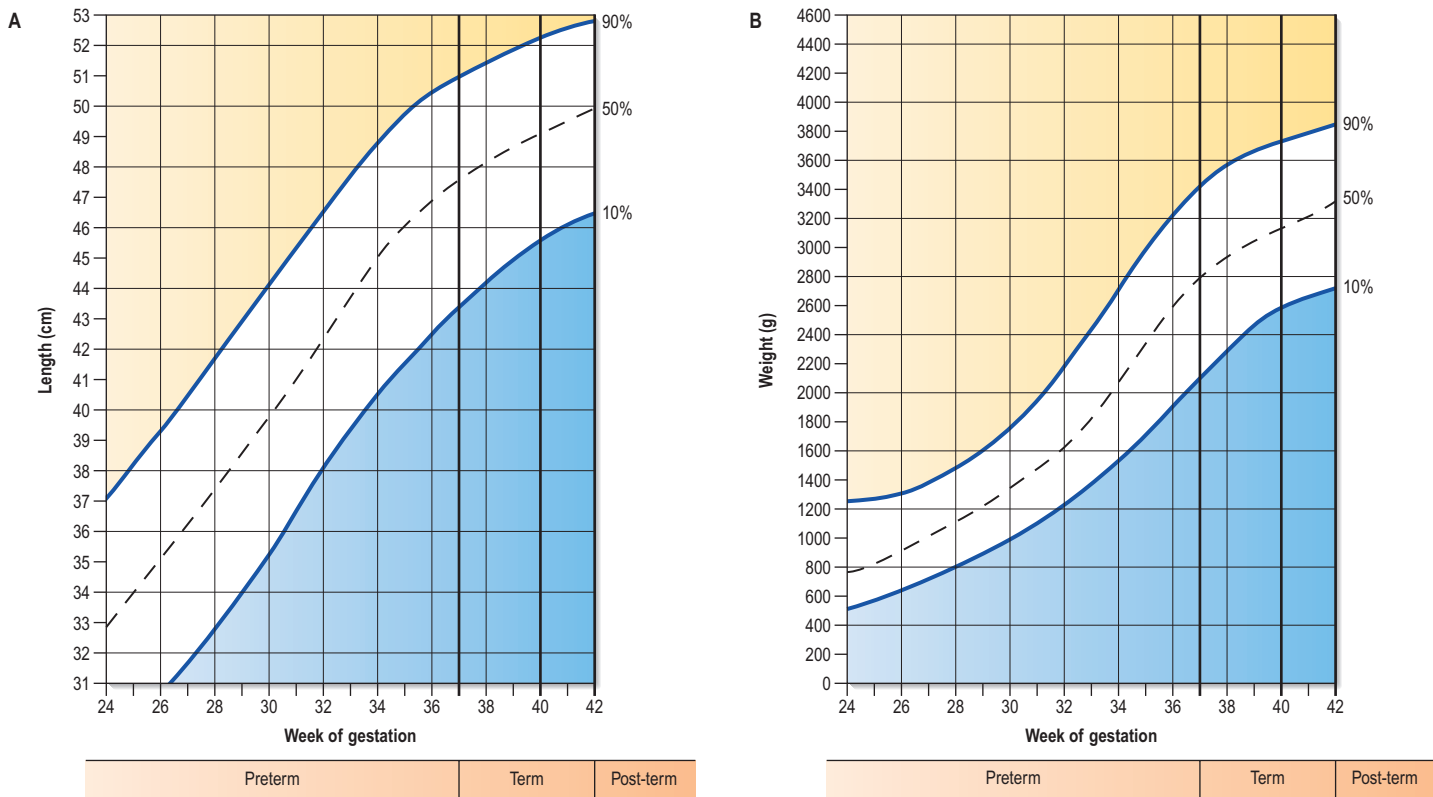


Fig. 14.9 Standardized graphs of (A) fetal length and (B) weight from 24 weeks of pregnancy, showing the tenth, fiftieth and ninetieth centiles.

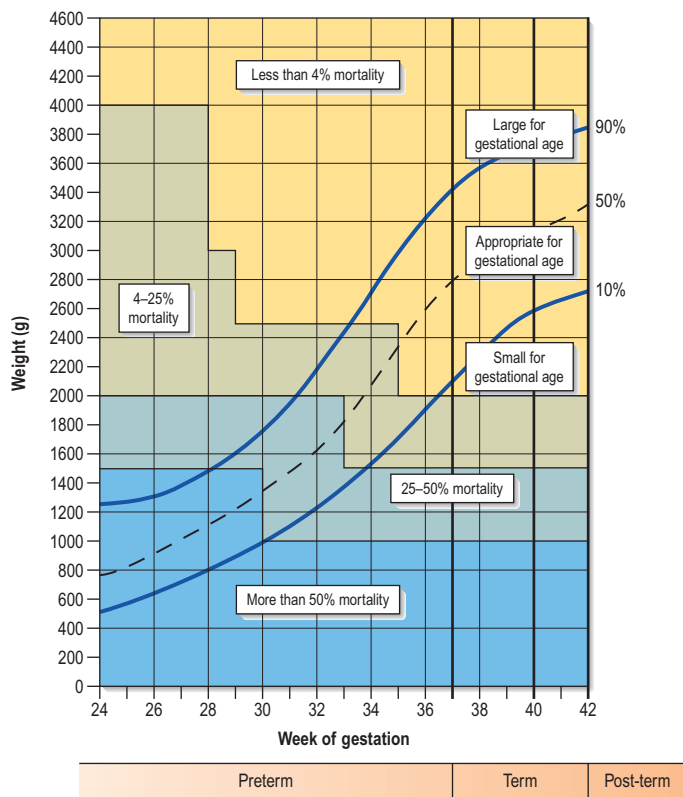


Fig. 14.10 Intrauterine growth status and its appropriateness for gestational age. Gestational age is more closely related to maturity than birth weight. The mortality for the weight ranges is indicated.

Growth rates of infants born prematurely differ from those born naturally at term. Normally, fetuses grow rapidly between 20 and 40 weeks. Optimum growth of premature infants is considered to be equivalent to intrauterine rates. However, whereas, *in utero*, fetuses show a slowing growth velocity before term, preterm infants show linear growth. Growth charts for preterm infants have been constructed (Fenton and Kim 2013). The INTERGROWTH-21st project is collecting data for preterm growth (see above).

Growth has always been regarded as a regular process. The rates of prenatal and postnatal growth can be indicated by increments in body length or weight, which, when plotted, form a growth curve (see Fig. 14.9). Growth curves can be plotted for individuals if accurate measurements are taken, preferably by the same person, for the entire period of growth, i.e. a longitudinal study. An alternative method is to collect a series of averages for each year of age obtained from different individuals, i.e. a cross-sectional study. Cross-sectional studies are valuable for the construction of standards for height and weight attained by healthy children at specific ages, and can establish centile limits of normal growth, but they cannot reveal individual differences in either the rate of growth or the timing of particular phases of growth.

The data from longitudinal and cross-sectional studies can also be used to plot the increments in height or weight from one age to the next. This produces a velocity curve, which reflects a child's state at any particular time much better than the growth curve, in which each point is dependent on the preceding one. The oldest published longitudinal study, still of great value today, was made by Count Philibert de Montbeillard on his son (Fig. 14.11). It shows that the velocity of growth in height decreases from birth onwards, and that a marked acceleration of growth, the adolescent growth spurt, occurs from 13 to 15 years (see below; see Fig. 14.14).

Cross-sectional data permit comparison of prenatal and postnatal growth. Childhood growth charts are used to predict normal childhood development. The velocity curve for the prenatal and postnatal period (Fig. 14.12) shows that the peak velocity for length is reached at about 4 months (note that these prenatal charts use the obstetric measurements of gestational time, in which fetal age is estimated from the last menstrual period, 2 weeks before fertilization). Growth in weight usually reaches its peak velocity after birth.

Tanner (in Harrison et al (1964)) noted that the more carefully measurements are taken, the more regular is the succession of points on a growth curve. However, a longitudinal study of growth measured weekly, semi-weekly and daily recorded that growth in length and head circumference occurred by saltatory increments, with a mean amplitude of 1.01 cm for length (Lampl 2002); growth stasis, steep changes in growth and incremental growth have all been recorded in infancy, childhood and adolescence (Caino et al 2006).

Charts of height and weight correlated to age are compiled from extensive cross-sectional growth studies. Such charts show the mean height or weight attained at each age, termed the fiftieth centile, and also the centile lines for the seventy-fifth, ninetieth and ninety-seventh centiles, in addition to the twenty-fifth, ninth and second centiles. The data shown in Figure 14.13 are derived from United Kingdom cross-sectional references. Any comparison of an individual growth curve

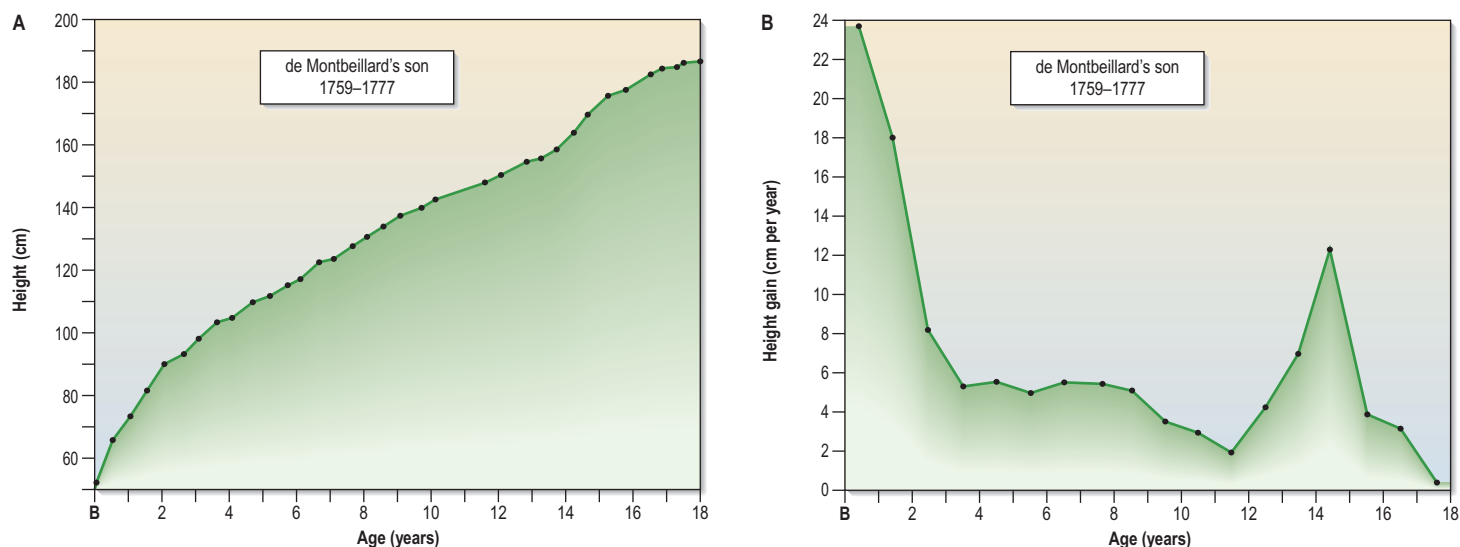


Fig. 14.11 A longitudinal study of growth. **A**, The height of de Montbeillard's son (1759–1777) from birth to 18 years. **B**, A growth velocity curve, plotting increments in height from year to year. (After D'Arcy Thompson, *On Growth and Form*, 1942, Cambridge University Press.)

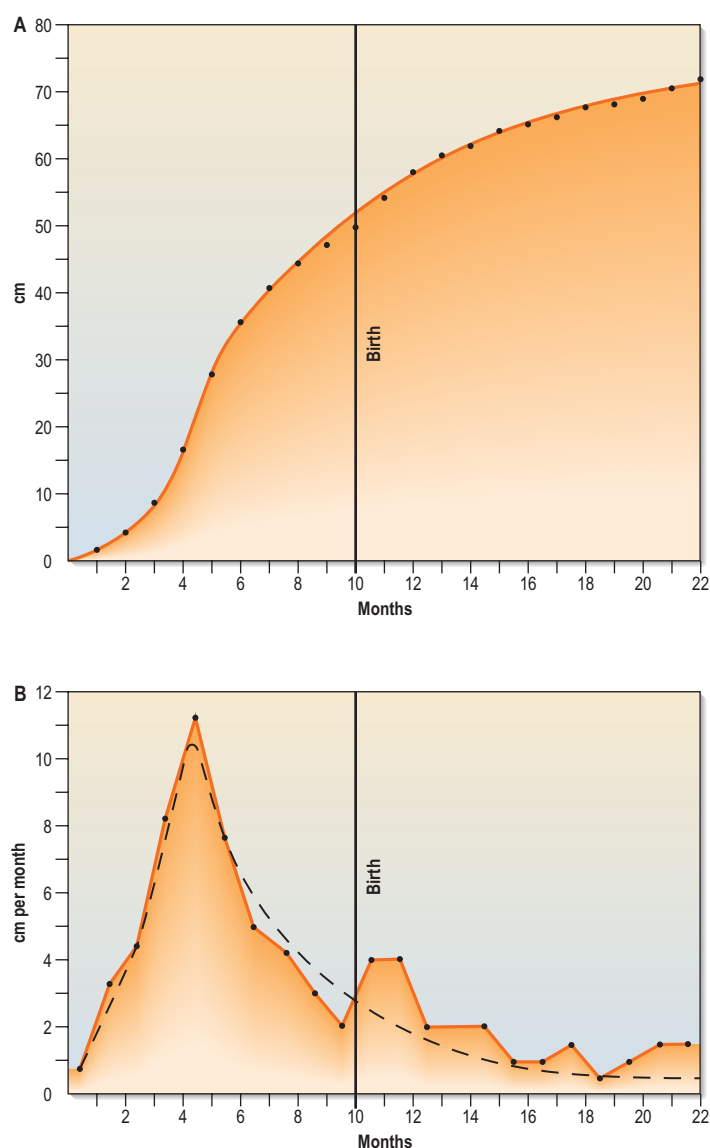


Fig. 14.12 Cross-sectional data showing growth. **A**, Growth in length in the prenatal and early postnatal periods. **B**, The corresponding velocity curve for this period. (After D'Arcy Thompson, *On Growth and Form*, 1942, Cambridge University Press.)

with these data must also take into account ethnicity and the nutritional and family history of that individual.

Plotting the growth of children on such charts provides guidelines for the prediction of normal growth and indicates when investigation of possible growth anomalies should occur. Children who grow in an environment that does not constrain their growth exhibit a pattern of growth that is mainly parallel to a particular centile, a phenomenon that has been termed homeorhesis or canalization (following the same imaginary 'canal' on the growth chart). After deviation from this centile as a consequence of the adolescent growth spurt, most children return to the same centile position in adulthood, a finding that suggests that this pattern is genetically determined within individuals (Cameron 2002). For a comprehensive account of all aspects of postnatal growth, the reader is directed to consult *Human Growth and Development* (Cameron 2002).

The WHO published growth charts for birth to 5 years and 5–19 years (2006). The charts were derived from children considered to be growing under optimal environmental conditions: they were breastfed, raised in environments that minimized poor diets and infection, and had non-smoking mothers. Similar growth charts have been published by WHO for specified regions (Mansourian et al 2012).

ADOLESCENT GROWTH SPURT AND ADULT SIZE

Growth charts reveal that body length increases from a neonatal range of 48–53 cm to about 75 cm during the first year after birth, and increases by 12–13 cm in the second year. Thereafter, 5–6 cm is added each year. In individual longitudinal growth curves, an increase in the velocity of growth occurs between 10.5 and 11 years in girls, and 12.5 and 13 years in boys. This rapid increase in growth is the adolescent growth spurt (Fig. 14.14; see Fig. 14.13). In both sexes, it lasts for 2–2.5 years. Girls gain about 16 cm in height during the spurt, with a peak velocity at 12 years of age. Boys gain about 20 cm in height (mostly by growth of the trunk), with a peak velocity at 14 years of age, during which time they may be growing at the rate of 10 cm a year.

Humans seem to be the only species to have a long quiescent interval between the rapid growth that takes place immediately after birth and the adolescent growth spurt. This type of growth has been termed the Infancy/Childhood/Puberty (ICP) growth model (Stevens et al 2013). It has been suggested that the quiescent interval allows the brain to mature, and complex social learning to take place, before individuals pass through puberty and become sexually active.

Apart from the expected change in growth velocity that occurs during puberty, it is now apparent that taller child height is prospectively associated with elevated risk of obesity (Johnson et al 2012, Van Dommelen 2014). Growth in height continues at a slower rate after the adolescent growth spurt. Noticeable growth is said to stop at 18 years in females and 20 years in males; longitudinal studies have indicated an average figure of 16.25 years (females) and 17.75 years (males) with a normal variation of ± 2 years (Harrison et al 1964). After this time, any increments that occur as a result of appositional growth at the cranial and caudal ends of the vertebral bodies and intervening intervertebral

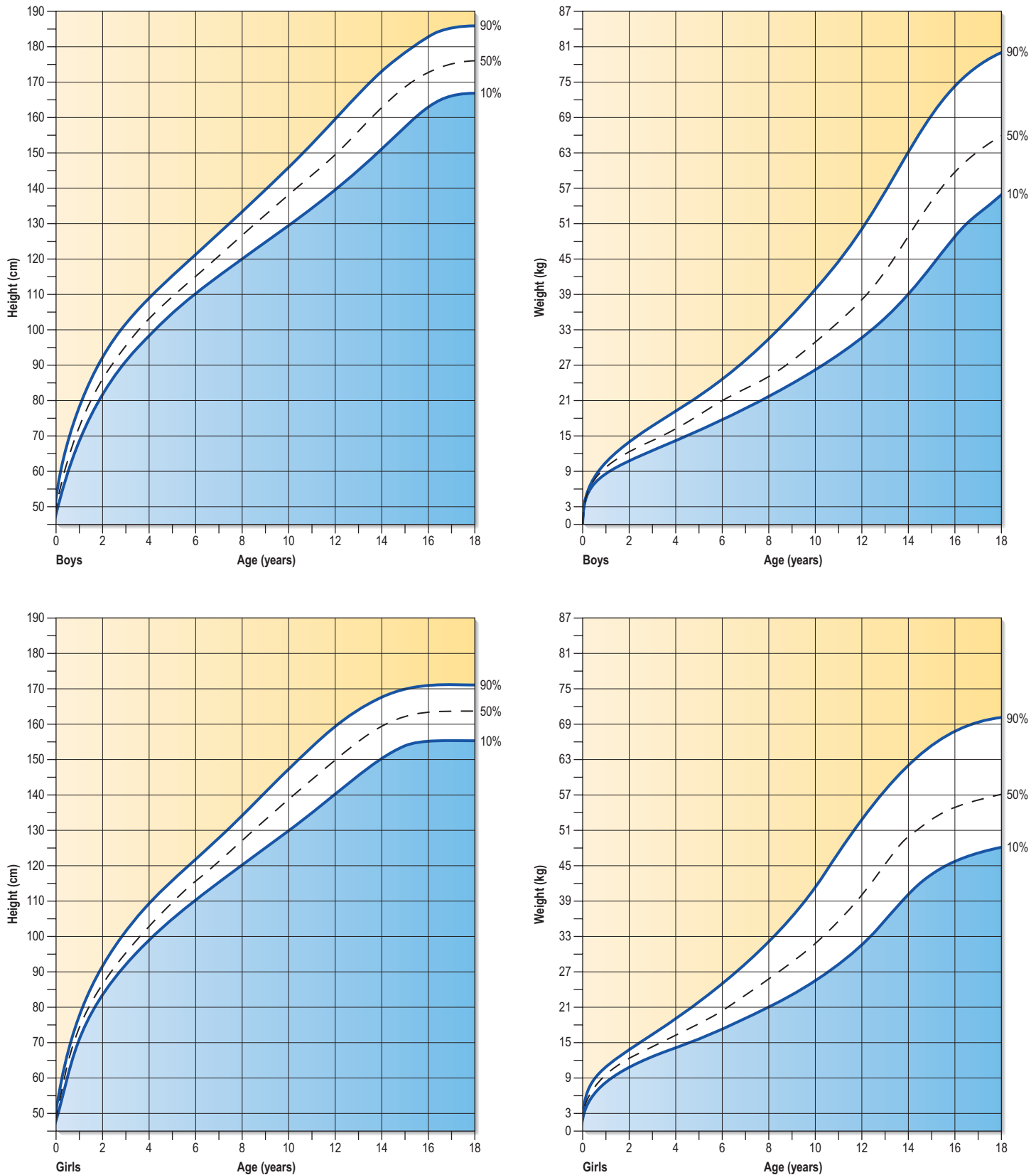


Fig. 14.13 Standard growth charts of British boys and girls showing the ninetieth, fiftieth and tenth centiles. (© Child Growth Foundation.)

discs are so small as to be difficult to measure. There is a loss of height after middle age.

Development-related gene expression correlates with three phases of human growth (infancy/early childhood; childhood/puberty; final height); the expression of clusters of growth-related, evolutionarily conserved genes varies in a development-dependent manner in human tissues (Stevens et al 2013). The outcomes of the Human Genome Project have been linked to anthropomorphic traits in meta-analyses of genome-wide association data that have identified genetic variants at hundreds of loci relevant to growth in pre-puberty, puberty and adulthood, particularly those associated with childhood and adult obesity (Genome-wide Investigation of ANthropometric measures – GIANT). A genome-wide association study has shown genetic loci linking

pubertal timing, height and growth with childhood obesity (Cousminer et al 2013).

The phenomenal growth rates of adolescence are most obvious in the increase in height. Weight gain is more variable. The birth weight is normally tripled by the end of the first year, and quadrupled by the end of the second year. Thereafter, weight increases by 2.25–2.75 kg annually until the adolescent growth spurt, when boys may add 20 kg to their weight and girls 16 kg. The peak velocity for weight gain lags behind the peak velocity for height by about 3 months. Body weight does not reach adult values until some time after adult height is attained. In recent years, the global estimates of the numbers of adults who are overweight or obese in adult life has exceeded the numbers who do not have sufficient food (Mascie-Taylor and Goto 2007).

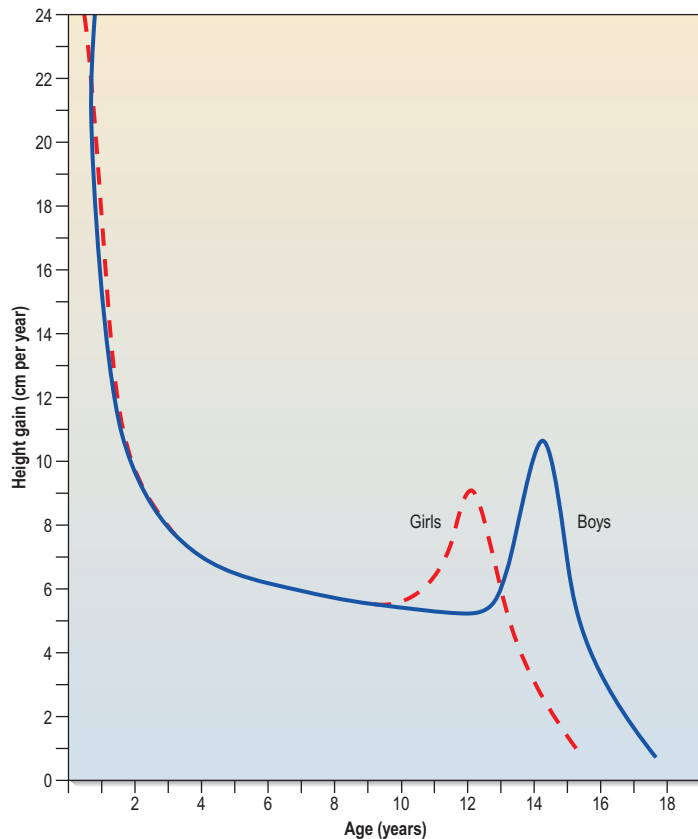


Fig. 14.14 Typical individual velocity curves for height: British boys and girls. (After Tanner JM, Whitehouse RH, Takaishi IM 1966 Standards from birth to maturity for height, weight, height velocity and weight velocity. Arch Dis Child 41:454–71, with permission from BMJ Publishing.)

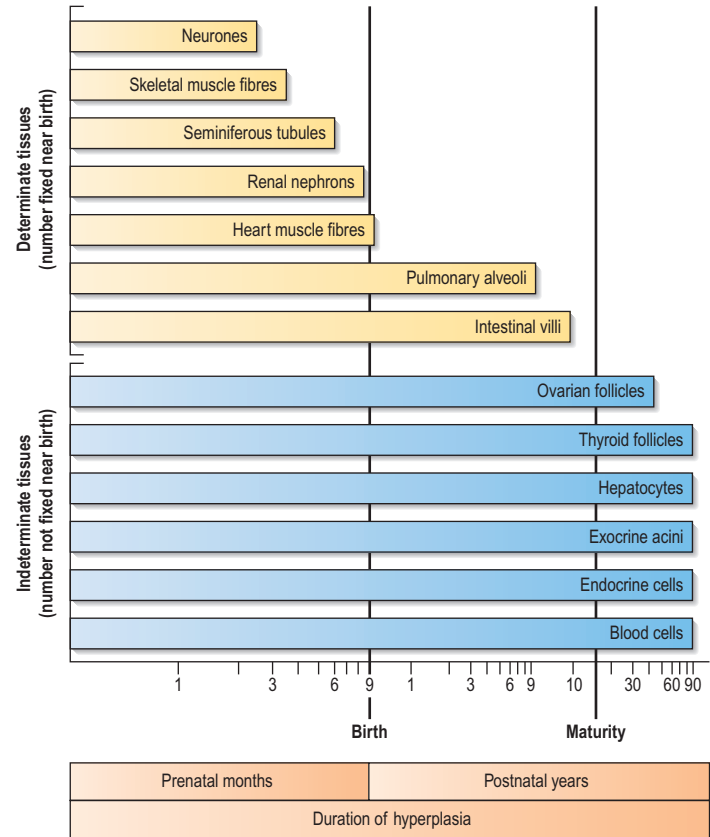


Fig. 14.15 The duration of multiplicative growth for various human tissues.

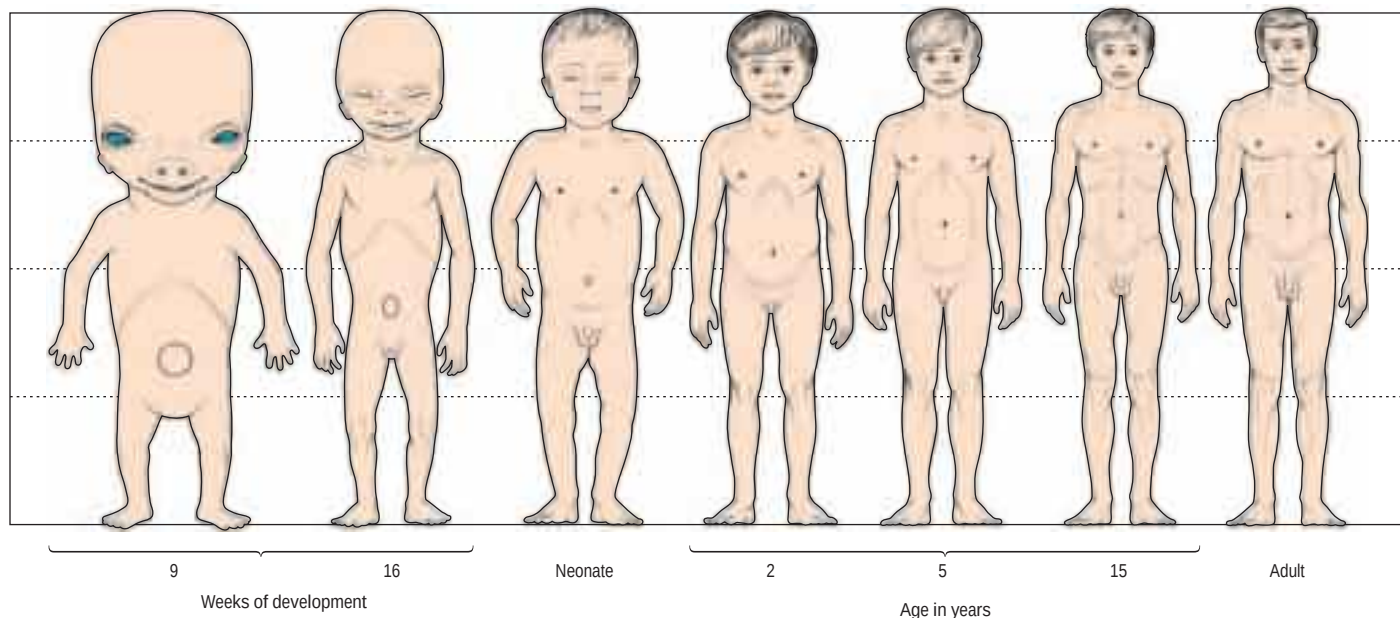


Fig. 14.16 Allometric growth in humans. The head is very large in proportion to the rest of the body during the embryonic period. After this time, the head grows more slowly than the torso and limbs and, by adulthood, the head is only one-eighth of the body length.

INTEGRATION OF TYPES OF GROWTH DURING DEVELOPMENT AND LIFE

In the later prenatal months and in the postnatal period, the various types of growth occur in differing patterns. The extent of tissue growth in organs depends on the specific duration of multiplicative growth for the cell types (Fig. 14.15). Different cell populations complete their initial developmental proliferation and become differentiated at different times; the final stage of differentiation is usually the cessation of cell division.

Growth of a body can be described in two ways: isometric and allometric. Isometric growth implies a progressive proportional increase in

all organs and systems with time; it does not occur in the developing embryo, which displays differential rates of growth. Allometric growth describes the differences in the relative rates of growth between one part of the body and another, and is most clearly seen in the changes in body proportion between fetuses, neonates, children and adults. Between 6 and 7 weeks after fertilization, the head is nearly one-half of the total embryonic length. It subsequently grows proportionally more slowly and, at birth, it is one-quarter of the entire length. During childhood, this pattern of growth continues with lengthening of the torso and limbs until, in adults, the head is one-eighth of the total length (Fig. 14.16).

Growth of the liver, spleen, kidneys, skeletal and muscular tissues generally follows pre- and postnatal growth curves given for the

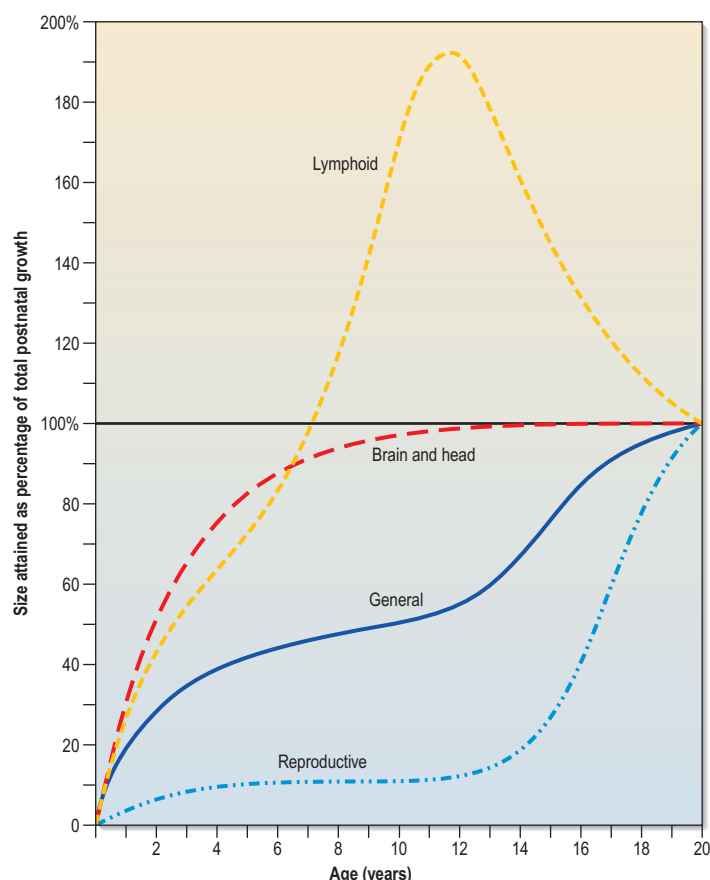


Fig. 14.17 Growth curves of different tissues, regions of the body and systems. Note that the growth of lymphoid tissue, thymus, lymph nodes and intestinal lymph masses decreases after puberty. (Adapted with permission from Tanner JM 1962 *Growth at Adolescence*, 2nd ed. Oxford: Blackwell Publishing.)

entire body. Other tissues have very different growth rates; the brain, skull, lymphoid tissues and reproductive organs all show differing growth rates during childhood and adolescence (Kappelou et al 2006) (Fig. 14.17).

Changes in growth at a tissue level are complex; differential growth rates during puberty can lead to temporal-specific associated pathology. For example, 30% of childhood fractures affect the radius, mostly the distal metaphysis, during the pubertal growth spurt, after which the incidence of these fractures decreases rapidly (Rauch 2012). Local differential rates of bone growth are seen in bone growth plates, peripheral and central trabecular bone, and in bone remodelling. During prepubertal growth, the distal radial growth plate adds about 9 mm to the length of the radius per year; the spaces between the peripheral trabeculae are filled with mineralized bone. It is suggested that incomplete trabecular coalescence during this rapid growth, rather than bone remodelling (which takes at least 6 months to complete), leads to lower bone mineral density, especially in boys, (which decreases rapidly after the pubertal growth spurt), and higher local metaphysial cortical porosity (Wang et al 2010, Rauch 2012).



Bonus e-book video

Video 14.1 Ultrasound features of the fetus at 26 weeks.

KEY REFERENCES

- Barker DJP, Osmond C, Simmonds SJ et al 1993 The relation of head size and thinness at birth to death from cardiovascular disease in adult life. *BMJ* 306:422–6.
A source of evidence for a link between birth weight and size and predictions of later cardiovascular pathology.
- Catalano PM, Hauguel-De Mouzon S 2011 Is it time to revisit the Pedersen hypothesis in the face of the obesity epidemic? *Am J Obstet Gynecol* 204:479–87.
This paper considers the changes in the demographics of obesity and its consequences for epigenetic changes to the feto-placental unit leading to generational chronic disease.
- Galan HL, Pandipati S, Filly RA 2008 Ultrasound evaluation of fetal biometry and normal and abnormal fetal growth. In: Callan PW (ed) *Ultrasonography in Obstetrics and Gynecology*. Philadelphia: Elsevier, Saunders; Ch. 7, pp. 225–65.
This chapter gives an overview of dating a pregnancy, measuring growth and assessing fetal status.
- Godfrey KM, Barker DJ 2000 Fetal nutrition and adult disease. *Am J Clin Nutr* 71:1344S–52S.
This paper considers the effect of maternal diet on the future risk factors for cardiovascular disease in her offspring.
- Harrison GA, Weiner JS, Tanner JM et al 1964 *Human Biology*. Oxford: Clarendon Press; Ch. 19.
A presentation of data on the human growth curve.
- Lampl M 2002 Saltation and stasis. In: Cameron N (ed) *Human Growth and Development*. New York: Academic Press; Ch. 12.
A presentation of the evidence for saltatory growth in humans.
- O’Rahilly R, Müller F 1987 *Developmental Stages in Human Embryos*. Washington: Carnegie Institution.
The most comprehensive and detailed morphological account of human development during the first 8 weeks of life; it sets out the basis for the staging system used for human embryos.
- O’Rahilly R, Müller F 2010 Developmental stages in human embryos: revised and new measurements. *Cells Tissues Organs* 192:73–84.
This paper presents the latest thoughts on matching stages to age and embryo length.
- Sepulveda W, Ximenes R, Wong AE et al 2012 Fetal magnetic resonance imaging and three-dimensional ultrasound in clinical practice: applications in prenatal diagnosis. *Best Pract Res Clin Obstet Gynaecol* 26: 593–624.
This paper gives examples of the clarity of 3D MRI.
- Uauy R, Casanello P, Krause B et al 2013 International Fetal and Newborn Growth Consortium for the 21st Century. Conceptual basis for prescriptive growth standards from conception to early childhood: present and future. *BJOG* 120 Suppl 2:3–8.
This paper is part of the INTERGROWTH-21st study, which aims to develop prescriptive intrauterine and newborn growth standards to establish what is meant by ‘normal’ fetal growth.

REFERENCES

- Adzick NS 2013 Prospects for fetal surgery. *Early Human Dev* 89:881–6.
- Barker DJP, Osmond C, Simmonds SJ et al 1993 The relation of head size and thinness at birth to death from cardiovascular disease in adult life. *BMJ* 306:422–6.
A source of evidence for a link between birth weight and size and predictions of later cardiovascular pathology.
- Bellini C, Boccardo F, Bonioli E et al 2006 Lymphodynamics in the fetus and newborn. *Lymphology* 39:110–17.
- Bhatta SRC, Keriakos R 2011 Review of the recent literature on the mode of delivery for singleton vertex preterm babies. *J Pregnancy* 2011:186560.
- Caino S, Kelmansky D, Adamo P et al 2006 Short-term growth in healthy infants, schoolchildren and adolescent girls. *Ann Hum Biol* 33: 213–26.
- Cameron N 2002 *Human Growth and Development*. New York: Academic Press.
- Canturk N, Atsu SS, Aka PS et al 2014 Neonatal line on fetus and infant teeth, an indicator of live birth and mode of delivery. *Early Human Dev* 90:393–7.
- Cass DL 2011 Impact of prenatal diagnosis and therapy on neonatal surgery. *Semin Fetal Neonatal Med* 16:130–8.
- Catalano PM, Hauguel-De Mouzon S 2011 Is it time to revisit the Pedersen hypothesis in the face of the obesity epidemic? *Am J Obstet Gynecol* 204:479–87.
This paper considers the changes in the demographics of obesity and its consequences for epigenetic changes to the feto-placental unit leading to generational chronic disease.
- Cheng P-J, Human S, Shaw SW et al 2010 Evaluation of fetal spine biometry between 11 and 14 weeks of gestation. *Ultrasound Med Biol* 36: 1060–5.
- Cilieborg MS, Boye M, Sangild PT 2012 Bacterial colonization and gut development in preterm neonates. *Early Human Dev* 88:541–9.
- Cohen AR, Couto J, Cummings JJ et al 2014 Position statement on fetal myelomeningocele repair. *Am J Obstet Gynecol* 210:107–11.
- Cousminer DL, Bery DJ, Timpson NJ et al 2013 Genome-wide association and longitudinal analyses reveal genetic loci linking pubertal height growth, pubertal timing and childhood adiposity. *Hum Mol Genet* 22:2735–47.
- Danzer E, Johnson MP 2014 Fetal surgery for neural tube defects. *Semin Fetal Neonatal Med* 19:2–8.
- Deprest J, Gratacos E, Nicolaides KH 2004 Fetoscopic tracheal occlusion (FETO) for severe congenital diaphragmatic hernia: evolution of a technique and preliminary results. *Ultrasound Obstet Gynecol* 24:121–6.
- Deprest J, Hodges R, Gratacos E et al 2014 Invasive fetal therapy. In: Creasy R, Resnik R, Iams JD et al (eds) *Creasy and Resnik's Maternal-Fetal Medicine: Principles and Practice*, 7th ed. Philadelphia: Elsevier, Saunders; Ch. 35, pp. 516–57.
- Fenton TR, Kim JH 2013 A systematic review and meta-analysis to revise the Fenton growth chart for preterm infants. *BMC Pediatrics* 13:59.
- Galan HL, Pandipati S, Filly RA 2008 Ultrasound evaluation of fetal biometry and normal and abnormal fetal growth. In: Callan PW (ed) *Ultrasonography in Obstetrics and Gynecology*. Philadelphia: Elsevier, Saunders; Ch. 7, pp. 225–65.
This chapter gives an overview of dating a pregnancy, measuring growth and assessing fetal status.
- Godfrey KM, Barker DJ 2000 Fetal nutrition and adult disease. *Am J Clin Nutr* 71:1344S–52S.
This paper considers the effect of maternal diet on the future risk factors for cardiovascular disease in her offspring.
- Harrison GA, Weiner JS, Tanner JM et al 1964 *Human Biology*. Oxford: Clarendon Press; Ch. 19.
A presentation of data on the human growth curve.
- Hyde MJ, Modi N 2012 The long-term effects of birth by caesarean section: the case for a randomised controlled trial. *Early Human Dev* 88: 943–9.
- Jakaitis BM, Denning PW 2014 Human breast milk and the gastrointestinal innate immune system. *Clin Perinatol* 41:423–35.
- Johnson W, Stovitz SD, Choh AC et al 2012 Patterns of linear growth and skeletal maturation from birth to 19 years of age in overweight young adults. *Int J Obes* 36:535–41.
- Kappelou O, Counsell SJ, Kennea N et al 2006 Abnormal cortical development after premature birth shown by altered allometric scaling of brain growth. *PLoS Med* 3:e265.
- Kelishadi R, Poursafa P 2014 A review on the genetic, environmental, and lifestyle aspects of the early-life origins of cardiovascular disease. *Curr Probl Pediatr Adolesc Health Care* 44:54–72.
- Lampl M 2002 Saltation and stasis. In: Cameron N (ed) *Human Growth and Development*. New York: Academic Press; Ch. 12.
A presentation of the evidence for saltatory growth in humans.
- Larma J, Landon MB 2011 Fetal weight: how large is too large? *Am J Obstet Gynecol* 204:457–8.
- Mansourian M, Marateb HR, Kelishadi R et al 2012 First growth curves based on the World Health Organization reference in a Nationally-Representative Sample of Pediatric Population in the Middle East and North Africa (MENA): the CASPIAN-III study. *BMC Pediatrics* 12:149.
- Mascie-Taylor CGN, Goto R 2007 Human variation and body mass index: a review of the universality of BMI cut-offs, gender and urban-rural differences, and secular changes. *J Physiol Anthropol* 26:109–12.
- Matamoros S, Gras-Leguen C, Le Vacon F et al 2013 Development of intestinal microbiota in infants and its impact on health. *Trends Microbiol* 21:167–73.
- McGuire W, Dyson L, Renfrew M 2010 Maternal obesity: consequences for children, challenges for clinicians and carers. *Semin Fetal Neonatal Med* 15:108–12.
- Merialdi M, Widmer M, Gülmezoglu AM et al 2014 WHO multicentre study for the development of growth standards from fetal life to childhood: the fetal component. *BMC Pregnancy Childbirth* 14:157.
- Neu J, Mai V 2012 The developing intestinal microbiome and its relationship to health and disease. In: Neu J *Gastroenterology and Nutrition: Neonatology Questions and Controversies*, 2nd ed. Philadelphia: Elsevier, Saunders; Ch. 5, pp. 59–65.
- Neu J, Rushing J 2011 Cesarean versus vaginal delivery: long term infant outcomes and the hygiene hypothesis. *Clin Perinatol* 38:321–31.
- O'Rahilly R, Müller F 1987 *Developmental Stages in Human Embryos*. Washington: Carnegie Institution.
The most comprehensive and detailed morphological account of human development during the first 8 weeks of life; it sets out the basis for the staging system used for human embryos.
- O'Rahilly R, Müller F 2000 Minireview: prenatal ages and stages – measures and errors. *Teratology* 61:382–4.
- O'Rahilly R, Müller F 2010 Developmental stages in human embryos: revised and new measurements. *Cells Tissues Organs* 192:73–84.
This paper presents the latest thoughts on matching stages to age and embryo length.
- Pedersen J 1952 Diabetes and pregnancy: blood sugar of newborn infants (PhD thesis). Copenhagen: Danish Science Press; p. 230.
- Plagemann A 2004 'Fetal programming' and 'functional teratogenesis': on epigenetic mechanisms and prevention of perinatally acquired lasting health risks. *J Perinat Med* 32:297–305.
- Poston L 2012 Maternal obesity, gestational weight gain and diet as determinants of offspring long term health. *Best Pract Res Clin Endocrinol Metab* 26:627–39.
- Pretorius DH, Aziz S, D'Agostini DA et al 2014 Clinical applications of three dimensional sonography in obstetrics. In: Creasy RK, Resnik R, Iams JD et al (eds) *Creasy and Resnik's Maternal-Fetal Medicine: Principles and Practice*, 7th ed. Philadelphia: Elsevier, Saunders; Ch. 19, pp. 218–233.
- Ramakrishna BS, Venkataraman S, Mukhopadhyaya A 2006 Tropical malabsorption. *Postgrad Med J* 82:779–87.
- Rauch 2012 The dynamics of bone structure development during pubertal growth. *J Musculoskelet Neuronal Interact* 12:1–6.
- Reddy UM, Abuhamad AZ, Levine D et al 2014 Fetal imaging: Executive Summary of a Joint Eunice Kennedy Shriver National Institute of Child Health and Human Development, Society for Maternal-Fetal Medicine, American Institute of Ultrasound in Medicine, American College of Obstetricians and Gynecologists, American College of Radiology, Society for Pediatric Radiology, and Society of Radiologists in Ultrasound Fetal Imaging Workshop. *Am J Obstet Gynecol* 210:387–97.
- Ross MG, Beal MH 2008 Adult sequelae of intrauterine growth restriction. *Semin Perinatol* 32:213–8.

PRE- AND POSTNATAL DEVELOPMENT

- Sabel N, Johansson C, Kühnisch J et al 2008 Neonatal lines in the enamel of primary teeth. A morphological and scanning electron microscopy investigation. *Arch Oral Biol* 53:954–63.
- Sarris I, Ioannou C, Ohuma EO et al 2013 Standardisation and quality control of ultrasound measurements taken in the INTERGROWTH-21st Project. *BJOG* 120 Suppl 2:33–7.
- Schwartz N, Quant HS, Sammel MD et al 2014 Macrosomia has its roots in early placental development. *Placenta* 35:684–90.
- Sepulveda W, Ximenes R, Wong AE et al 2012 Fetal magnetic resonance imaging and three-dimensional ultrasound in clinical practice: applications in prenatal diagnosis. *Best Pract Res Clin Obstet Gynaecol* 26: 593–624.
This paper gives examples of the clarity of 3D MRI.
- Song SJ, Dominquez-Bello MG, Knight R 2013 How delivery mode and feeding can shape the bacterial community in the infant gut. *CMAJ* 185:373–4.
- Steer PJ, Modi N 2009 Elective caesarean sections – risks to the infant. *Lancet* 374:675–6.
- Stevens A, Hanson D, Whatmore A et al 2013 Human growth is associated with distinct patterns of gene expression in evolutionarily conserved networks. *BMC Genomics* 14:547.
- Streeter GL 1942 Developmental horizons in human embryos. Descriptions of age group XI, 13 to 20 somites, and age group XII, 21 to 29 somites. *Contrib Embryol Carnegie Inst Washington* 30:211–45.
- Streeter GL 1945 Developmental horizons in human embryos. Description of age group XIII, embryos of about 4 or 5 millimeters long, and age group XIV, period of indentation of the lens vesicle. *Contrib Embryol Carnegie Inst Washington* 31:27–63.
- Streeter GL 1948 Developmental horizons in human embryos. Description of age groups XV, XVI, XVII, and XVIII, being the third issue of a survey of the Carnegie collection. *Contrib Embryol Carnegie Inst Washington* 32:133–203.
- Uauy R, Casanello P, Krause B et al 2013 International Fetal and Newborn Growth Consortium for the 21st Century. Conceptual basis for prescriptive growth standards from conception to early childhood: present and future. *BJOG* 120 Suppl 2:3–8.
This paper is part of the INTERGROWTH-21st study, which aims to develop prescriptive intrauterine and newborn growth standards to establish what is meant by 'normal' fetal growth.
- Valentine CJ, Morrow AL 2012 Human milk feeding of the high risk neonate. In: *Neu J Gastroenterology and Nutrition: Neonatology Questions and Controversies*, 2nd ed. Philadelphia: Elsevier, Saunders; Ch.13, pp. 203–12.
- van Dommelen P, de Kroon MLA, Cameron N et al 2014 The impact of height during childhood on the national prevalence rates of overweight. *PLoS ONE* 9:e85769.
- Vrachnis N, Antonakopoulos N, Iliodromiti Z et al 2012 Impact of maternal diabetes on epigenetic modifications leading to diseases in the offspring. *Exp Diabetes Res* 2012:538474.
- Walsh JM, McAuliffe FM 2012 Prediction and prevention of the macrosomic fetus. *Euro J Obstet Gynecol Reprod Biol* 162:125–30.
- Wang Q, Wang XF, Iuliano-Burns S et al 2010 Rapid growth produces transient cortical weakness: a risk factor for metaphyseal fractures during puberty. *J Bone Miner Res* 25:1521–6.
- Werner EF, Savitz DA, Janevic TM et al 2012 Mode of delivery and neonatal outcomes in preterm, small-for-gestational-age newborns. *Obstet Gynecol* 120:560–4.
- WHO Multicentre Growth Reference Study Group 2006 WHO child growth standards based on length/height, weight and age. *Acta Paediatr Suppl* 450:76–85.
- Wynn JL, Neu J 2012 The neonatal gastrointestinal tract as a conduit to systemic inflammation and developmental delays. In: *Neu J Gastroenterology and Nutrition: Neonatology Questions and Controversies*, 2nd ed. Philadelphia: Elsevier, Saunders; Ch 19, pp. 293–304.
- Yang H, Kramer MS, Platt RW et al 2002 How does early ultrasound scan estimation of gestational age lead to higher rates of preterm birth? *Am J Obstet Gynecol* 186:433–7.